

FRAUD DETECTION SYSTEM

Major Project Report

Credit Card Fraud Detection Using Machine Learning

Submitted By: _____

Course/Branch: _____

Year/Semester: _____

Roll No.: _____

College: _____

Academic Session: 2026-27

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ABSTRACT

Credit-card fraud is a major financial and security problem. A fraud detection system must identify suspicious transactions while minimizing false alarms.

This project develops a machine-learning based Fraud Detection System using the ULB Credit Card Fraud Detection dataset. The project studies both supervised classification and unsupervised anomaly detection.

Logistic Regression and Random Forest are used as supervised classification models. Because fraudulent transactions are extremely rare compared with legitimate transactions, the project uses imbalance-aware class weighting, stratified splitting and threshold tuning. Performance is evaluated using precision, recall, F1-score, ROC-AUC and Precision-Recall AUC (PR-AUC), rather than relying only on accuracy.

Isolation Forest is additionally implemented as an anomaly-detection method.

The final system produces fraud probabilities, fraud predictions and business risk levels (Low, Medium, High and Critical). The project demonstrates the complete machine-learning workflow from data loading and exploratory analysis to model training, evaluation and risk interpretation.

1. INTRODUCTION

Digital payments have increased the importance of automatic transaction monitoring. Fraudulent transactions can cause direct financial losses, chargebacks, customer dissatisfaction and reputational damage.

Manual checking of every transaction is not practical at large scale.

Machine learning can learn patterns from historical transactions and assign a probability or risk score to new transactions.

The selected ULB/Kaggle dataset contains anonymized credit-card transactions. It has 284,807 transactions, of which 492 are fraudulent. Fraud therefore represents only about 0.172% of the transactions. The features V1 to V28 are PCA-transformed numerical variables; Time and Amount are also provided, and Class is the target variable.

2. PROBLEM STATEMENT

Fraudulent activities cause major financial losses. The goal is to develop a system that can identify fraudulent transactions using machine-learning techniques while handling severe class imbalance.

3. OBJECTIVES

1. Load and understand the credit-card transaction dataset.
2. Perform exploratory data analysis.
3. Handle the highly imbalanced target class.
4. Train supervised classification models.
5. Apply anomaly detection.
6. Tune the classification threshold.
7. Evaluate fraud-detection performance.
8. Generate transaction risk levels.
9. Interpret the results from a business-risk perspective.

4. DATASET DESCRIPTION

Dataset:

Credit Card Fraud Detection - ULB / Kaggle

Source:

<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

Important columns:

- Time: seconds elapsed between the transaction and the first transaction.
- V1 to V28: anonymized PCA-transformed numerical features.
- Amount: transaction amount.
- Class: target; 0 = legitimate, 1 = fraud.

Dataset characteristics:

- Total transactions: 284,807
- Fraudulent transactions: 492
- Legitimate transactions: 284,315
- Fraud rate: approximately 0.172%

The target class is severely imbalanced. This is the central machine-learning challenge in the project.

5. TECHNOLOGIES USED

Programming Language:

Python

Libraries:

- Pandas
- NumPy
- Matplotlib

- Scikit-learn
- Joblib
- Jupyter Notebook (optional)

Machine-learning techniques:

- Logistic Regression
- Random Forest Classification
- Isolation Forest Anomaly Detection

6. SYSTEM WORKFLOW

Dataset

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Data Loading

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v

Data Validation

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v

Exploratory Data Analysis

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v

Train / Validation / Test Split

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v

Preprocessing + Scaling

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v

Imbalance Handling

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+-----+

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v v

Logistic Regression Random Forest

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+-----+-----+

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v

Threshold Tuning

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v

Final Evaluation

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v

Fraud Probability / Risk

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v

Business Decision

Parallel:

Isolation Forest -> Anomaly Score -> Fraud Alert

7. DATA PREPROCESSING

The dataset is loaded from CSV and checked for required columns and valid target labels.

The target column `Class` is separated from the input features.

The data is divided into:

- Training set: 64%
- Validation set: 16%
- Test set: 20%

The split is stratified so that the rare fraud class remains represented in each subset.

For Logistic Regression, numerical features are imputed using the training median when necessary and standardized using StandardScaler. The scaler is fit only on training data and then applied to validation/test data. This prevents data leakage.

8. HANDLING IMBALANCED DATA

The fraud class is much smaller than the legitimate class. A model that predicts every transaction as legitimate could achieve very high accuracy while detecting no fraud.

To address this:

- Logistic Regression uses `class\_weight='balanced'`.
- Random Forest uses `class\_weight='balanced\_subsample'`.
- Stratified train/validation/test splits are used.
- The decision threshold is tuned on the validation set.
- Precision, recall, F1 and PR-AUC are emphasized.

PR-AUC is particularly informative for rare positive classes because it focuses on the precision-recall trade-off for fraud detection.

9. MACHINE LEARNING MODELS

9.1 Logistic Regression

Logistic Regression estimates the probability that a transaction belongs to

the fraud class.

Advantages:

- Simple and fast.
- Easy to interpret.
- Provides probabilities.
- Useful as a baseline model.

9.2 Random Forest

Random Forest combines many decision trees and averages their predictions.

It can learn nonlinear relationships and interactions between features.

Advantages:

- Handles nonlinear patterns.
- Usually strong on tabular data.
- Provides probability estimates.
- Less sensitive to feature scaling than Logistic Regression.

9.3 Isolation Forest

Isolation Forest is an unsupervised anomaly-detection method. It attempts to isolate unusual observations quickly. Transactions identified as unusual are treated as anomaly/fraud alerts.

This model is useful because in a real financial system new fraud patterns may appear that were not present in the historical labels.

10. THRESHOLD TUNING

A probability of 0.50 is not always the best decision boundary for fraud detection.

The project checks thresholds from 0.05 to 0.95 on the validation set and selects the threshold that maximizes F1-score.

This threshold is then applied once to the untouched test set.

The advantage is that the project explicitly balances precision and recall instead of assuming that 0.50 is optimal.

11. EVALUATION METRICS

Precision:

Of all transactions predicted as fraud, how many were actually fraud?

Recall:

Of all actual fraud transactions, how many were detected?

F1-score:

Harmonic mean of precision and recall.

ROC-AUC:

Measures ranking performance across classification thresholds.

PR-AUC:

Measures the precision-recall trade-off and is especially useful for highly imbalanced fraud-detection problems.

Confusion Matrix:

Shows true negatives, false positives, false negatives and true positives.

For fraud detection, false negatives can be particularly costly because a fraudulent transaction may be missed.

12. BUSINESS RISK INTERPRETATION

The system converts predicted fraud probabilities into four risk levels:

Low : probability < 0.20

Medium : 0.20 to < 0.50

High : 0.50 to < 0.80

Critical : probability ≥ 0.80

These levels are illustrative and should be calibrated using real business costs before deployment.

A practical system could:

- Allow low-risk transactions automatically.
- Apply additional verification to medium-risk transactions.
- Send high-risk transactions for review.
- Temporarily block or strongly verify critical-risk transactions.

13. RESULTS

After running ``python fraud_detection.py``, the following files are generated:

1. outputs/metrics.json
2. outputs/fraud_predictions.csv
3. outputs/class_distribution.png
4. outputs/amount_distribution.png
5. outputs/logistic_regression_confusion_matrix.png
6. outputs/random_forest_confusion_matrix.png
7. outputs/roc_curves.png
8. outputs/precision_recall_curves.png

IMPORTANT:

The numerical model results should be copied from ``outputs/metrics.json`` after the student runs the project on the downloaded dataset. No performance numbers are fabricated in this report.

Results table to complete:

Model	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	_____	_____	_____	_____	_____
Random Forest	_____	_____	_____	_____	_____

| Isolation Forest | _____ | _____ | _____ | N/A | N/A |

Best supervised model:

Selected validation threshold:

14. ADVANTAGES

1. Handles severe class imbalance.
2. Uses multiple machine-learning approaches.
3. Uses validation-based threshold tuning.
4. Uses fraud-focused evaluation metrics.
5. Produces transaction-level risk levels.
6. Includes both supervised and unsupervised approaches.
7. Can be extended to real-time transaction scoring.

15. LIMITATIONS

1. The dataset is anonymized, so business meaning of V1-V28 is unavailable.
2. The dataset represents historical transactions and may not represent current fraud behavior.
3. A model trained on this dataset should not be directly deployed in a real bank without additional validation.
4. Thresholds should be selected using actual financial costs.
5. Fraud patterns can change over time, requiring model monitoring and retraining.
6. Isolation Forest is an anomaly detector, not proof that a transaction is fraudulent.

16. FUTURE SCOPE

1. Add XGBoost or LightGBM for stronger tabular classification.
2. Use cost-sensitive learning based on actual fraud losses.
3. Add real-time streaming transaction detection.
4. Build a Flask/FastAPI web application.
5. Create a dashboard using Streamlit or Power BI.
6. Add model explainability using SHAP.
7. Monitor model drift.
8. Retrain the model periodically with recent transactions.
9. Combine supervised probability with anomaly score.
10. Add human-review feedback to improve the model.

17. CONCLUSION

The project demonstrates a complete machine-learning pipeline for credit-card

fraud detection. The main challenge is severe class imbalance, which makes accuracy an inadequate standalone metric. The project addresses this through imbalance-aware models, stratified data splitting, validation-based threshold tuning and fraud-focused metrics.

Logistic Regression provides an interpretable baseline, Random Forest captures nonlinear patterns, and Isolation Forest provides an unsupervised anomaly detection approach. The final system produces fraud predictions and risk levels that can support transaction review.

The project provides practical experience in Python, Pandas, Scikit-learn, data preprocessing, imbalanced classification, anomaly detection, model evaluation and business-risk interpretation.

18. REFERENCES

1. Kaggle - Credit Card Fraud Detection:

<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

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3. Dal Pozzolo et al., "Learned lessons in credit card fraud detection from a practitioner perspective", Expert Systems with Applications, 2014.

4. Scikit-learn documentation:

<https://scikit-learn.org/>

END OF REPORT